Project Design Phase-II

Solution Requirements(Functional& Non-functional)

Date	31 January 3035
Team ID	SWTID-2026-8933
Project Name	Script-Controlled ACL Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Record Access Control	Configure a READ Access Control List (ACL) for the target table.
FR-2	User Validation	Identify the currently logged-in user requesting access to the record.
FR-3	Field Value Validation	Retrieve and validate the specified field value from the record.
FR-4	Script-Based Access Decision	Execute a custom script to compare the user information with the specified record field value.
FR-5	Access Permission	Allow the user to read the record when the defined access condition is satisfied.
FR-6	Access Restriction	Deny read access when the defined condition is not satisfied.

FR-7	Condition Handling	Handle different field values and user conditions according to the access rules.
FR-8	Access Verification	Test the ACL using authorized and unauthorized users to verify the access behavior.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The access control configuration should be simple to understand, manage, and maintain.
NFR-2	Security	The solution must prevent unauthorized users from reading restricted records.
NFR-3	Reliability	The ACL should consistently provide the correct access decision based on the defined conditions.
NFR-4	Performance	The access validation script should execute efficiently without causing unnecessary delays while accessing records.
NFR-5	Availability	The access control mechanism should function whenever users attempt to access the protected records.
NFR-6	Scalability	The solution should support additional access conditions and records without requiring major changes to the overall design.
NFR-7	Maintainability	The script and access rules should be structured so that future modifications can be made easily.
NFR-8	Auditability	Access control logic should be clearly defined so that the reason for allowing or denying access can be verified during testing and maintenance.